

Pump and probe response in non-linear optics language

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Notation

In the following, we assume that

$$E(t) = \int_0^\infty d\omega E_\omega \cos(\omega t) \quad (1)$$

which implies $E_\omega = E_{-\omega}$ with $E_\omega \in \mathbb{R}$. This is slightly less general than the case $E_\omega = E_{-\omega}^*$ with $E_\omega \in \mathbb{C}$

PUMP AND PROBE FROM NON-LINEAR OPTICS COEFFICIENTS

One can define an effective response function that is measured in the experiments as:

$$\chi^{eff}(\omega) = \frac{P(\omega)}{E_p(\omega)} \quad (2)$$

where E_p is the probe field, and $P(t)$ is the time dependent polarization induced by the probe field. For weak probes $\chi_{eff}(\omega)$ is the first order response function.

$$\chi^{eff}(\omega) = \chi^{eff,(1)}(\omega) + O(E_p^2) \quad (3)$$

Equilibrium case

If we probe a system at equilibrium, $P(t) = P_p(t)$, where $P_p(t)$ is the results of a real-time propagation in presence of the probe only. Then, $\chi^{eff,(1)}(\omega) = \chi^{eq,(1)}(\omega)$ is the equilibrium response function.

Pump and probe experiments

In presence of a pump field E_P , we consider $P(t) = P_{Pp}(t) - P_P(t)$, where $P_{Pp}(t)$ is the results of a real-time propagation in presence of both the pump and the probe field, while $P_P(t)$ is the result of a real time simulation in presence of the pump only. Then, $\chi^{eff,(1)}(\omega) = \chi^{neq,(1)}[E_P](\omega)$ is a non equilibrium response function that we use to compute non equilibrium spectra:

$$\chi^{neq,(1)}[E_P](\omega) = \frac{P_{Pp}(\omega) - P_P(\omega)}{E_p(\omega)} \quad (4)$$

Such response function in general depends on the pump field. In presence of pulsed fields, it also depends on the time delay τ between the pump and the probe, e.g. we can write $\chi^{neq,(1)}[E_P](\omega, \tau)$ which is usually studied in pump and probe spectroscopy measurements. If the pump can be approximated with an ever-lasting field, or in general if no dynamics is involved beyond the coherent induced oscillations (e.g. no dissipative mechanisms are accounted for), this τ dependence reduces to the distance between the probe time and a phase maximum of the pump, e.g. the phase dealy.

Finally, in case also the probe can be approximated as ever-lasting, we can define a response function $\chi^{neq,(1)}[E_P](\omega)$ independent of τ .

Non linear optics in presence of two fields

We label the two fields as pump E_P and probe E_p . We consider $P(t) = P_{Pp}(t)$ and define

$$\chi^{nl,(1)}[E_P](\omega) = \frac{P_{Pp}(\omega)}{E_p(\omega)}. \quad (5)$$

This can be expressed in terms of higher order equilibrium response functions w.r.t. both the pump and the probe, e.g. to both the fields. We assume now a mono-cromatic pump and label ω_P the pump frequency, e.g. $E_P(\omega) = \delta(\omega - \omega_P)E_{\omega_P}$. Then, we write the polarization in terms of the response functions keeping the terms up to 1st order in the probe and 2nd order in the pump

N.B.: the non linear response function are usually written with $(n + 1)$ frequencies, when n frequencies come from the fields and the extra frequency is the one of the polarization. To be fixed in the equations below.

$$\begin{aligned}
P_{Pp}(\omega) = & \chi^{eq,eff}(\omega)E_P(\omega) + \{\chi^{eq,(1)}(\omega)E_p(\omega) \\
& + [\chi^{eq,(2)}(\omega - \omega_P, \omega_P)E_p(\omega - \omega_P) + \chi^{eq,(2)}(\omega + \omega_P, -\omega_P)E_p(\omega + \omega_P)]E_{\omega_P} \\
& + [\chi^{eq,(3)}(\omega, \omega_P, -\omega_P)E_p(\omega) \\
& + \chi^{eq,(3)}(\omega - 2\omega_P, \omega_P, \omega_P)E_p(\omega - 2\omega_P) + \chi^{eq,(3)}(\omega + 2\omega_P, -\omega_P, -\omega_P)E_p(\omega + 2\omega_P)]E_{\omega_P}^2 \\
& O(E_P^3)\} \quad (6)
\end{aligned}$$

Dividing by $E_p(\omega)$ we obtain the response function

$$\chi^{nl,(1)}[E_P, \omega_P](\omega) = \chi^{eq,eff}(\omega) \frac{E_P(\omega)}{E_p(\omega)} + \frac{\{\dots\}}{E_p(\omega)} \quad (7)$$

The zero order (in the pump) term

The first term on the r.h.s., which is zero unless $\omega = \omega_P$, is also what we subtract when defining $\chi^{neq,(1)}(\omega)$. Indeed

$$\chi^{neq,(1)}[E_P, \omega_P](\omega) = \chi^{nl,(1)}(\omega) - \frac{P_P(\omega)}{E_p(\omega)} \quad (8)$$

$$= \chi^{neq,(1)}(\omega) - \frac{P_P(\omega)}{E_P(\omega)} \frac{E_P(\omega)}{E_p(\omega)} \quad (9)$$

$$= \chi^{neq,(1)}(\omega) - \chi^{eff,eq}(\omega) \frac{E_P(\omega)}{E_p(\omega)} \quad (10)$$

The final result

Using this, we finally obtain

$$\begin{aligned}
\chi^{neq,(1)}[E_P, \omega_P](\omega) - \chi^{eq,(1)}(\omega) = & \\
& \left[\chi^{eq,(2)}(\omega + \omega_P, -\omega_P) \frac{E_p(\omega + \omega_P)}{E_p(\omega)} + \chi^{eq,(2)}(\omega - \omega_P, \omega_P) \frac{E_p(\omega - \omega_P)}{E_p(\omega)} \right] E_P(\omega_P) + \\
& [\chi^{eq,(3)}(\omega, \omega_P, -\omega_P) \\
& + \chi^{eq,(3)}(\omega - 2\omega_P, \omega_P, \omega_P) \frac{E_p(\omega - 2\omega_P)}{E_p(\omega)} + \chi^{eq,(3)}(\omega + 2\omega_P, -\omega_P, -\omega_P) \frac{E_p(\omega + 2\omega_P)}{E_p(\omega)}] E_P^2(\omega_P) + \\
& O(E_P^3) \quad (11)
\end{aligned}$$

Two simplifications can be obtained:

- by considering an instantaneous probe, e.g. $E_p(t) = \delta(t)E_0$, for which $E_p(\omega) = E_0$:

$$\begin{aligned}
\chi^{neq,(1)}[E_P, \omega_P](\omega) - \chi^{eq,(1)}(\omega) = \\
\left[\chi^{eq,(2)}(\omega + \omega_P, -\omega_P) + \chi^{eq,(2)}(\omega - \omega_P, \omega_P) \right] E_{\omega_P} + \\
\left[\chi^{eq,(3)}(\omega, \omega_P, -\omega_P) + \chi^{eq,(3)}(\omega - 2\omega_P, \omega_P, \omega_P) + \chi^{eq,(3)}(\omega + 2\omega_P, -\omega_P, -\omega_P) \right] E_{\omega_P}^2 + \\
O(E_{\omega_P}^3) \quad (12)
\end{aligned}$$

- In case the probe field is monochromatic:

$$\chi^{neq,(1)}[E_P, \omega_P](\omega) - \chi^{eq,(1)}(\omega) = \chi^{eq,(3)}(\omega, \omega_P, -\omega_P) E_{\omega_P}^2 + O(E_{\omega_P}^4) \quad (13)$$

Otherwise, for broad pulsed probes, one needs to take into account the shape of the probe.

Revised analysis

We write again the previous formula in order to fix some notation and understand better the main result. The general starting point is the expansion of the polarization in frequency domain in terms of the equilibrium non-linear χ .

$$\begin{aligned}
P_i^{(1)}(\omega) &= \sum_{jn} \chi_{ij}^{(1)}(\omega; \omega_n) E_j(\omega_n) \\
P_i^{(2)}(\omega) &= \sum_{jknm} \chi_{ijk}^{(2)}(\omega; \omega_n, \omega_m) E_j(\omega_n) E_k(\omega_m) \quad (14)
\end{aligned}$$

where the condition $\omega = \omega_n$ hold for $\chi^{(1)}$, while $\omega = \omega_n + \omega_m$ holds for the $\chi^{(2)}$ and so on so forth for the higher terms. Note that, if one is interested P at a given ω value then the sum over $n, m, \dots$ are not considered and we need to understand the possible combination of the fields energies that can produce the wanted ω value.

Next, we adopt a two field representation splitting the monochromatic pump $E_P(\omega_P)$ (with fixed frequency $\pm\omega_P$) and a weak probe $E_p(\omega_p)$ with ω_p in a finite energy window. Using this representation for the field, we can write the polarization up to the third order. We omit the vector indexes and the relative sums and we choose to name the various ω in a way consistent with the fact the the sum of the frequency of the fields corresponds to the ω of

the polarization

$$\begin{aligned}
P^{(1)}(\omega) &= \chi^{(1)}(\omega; \omega_P) E_P(\omega_P) \delta(\omega - \omega_P) + \chi^{(1)}(\omega; \omega) E_p(\omega) \\
P^{(2)}(\omega) &= \chi^{(2)}(\omega; \omega_P, \omega_P) E_P(\omega_P) E_P(\omega_P) \delta(\omega - 2\omega_P) + \\
&\quad + \chi^{(2)}(\omega; \omega_P, \omega - \omega_P) E_P(\omega_P) E_p(\omega - \omega_P) + \\
&\quad + \chi^{(2)}(\omega; -\omega_P, \omega + \omega_P) E_P(\omega_P) E_p(\omega + \omega_P) \\
P^{(3)}(\omega) &= \chi^{(3)}(\omega; \omega_P, \omega_P, \omega_P) E_P(\omega_P) E_P(\omega_P) E_P(\omega_P) \delta(\omega - 3\omega_P) + \\
&\quad + \chi^{(3)}(\omega; \omega_P, \omega_P, \omega - 2\omega_P) E_P(\omega_P) E_P(\omega_P) E_p(\omega - 2\omega_P) \\
&\quad + \chi^{(3)}(\omega; \omega_P, -\omega_P, \omega) E_P(\omega_P) E_P(\omega_P) E_p(\omega) \\
&\quad + \chi^{(3)}(\omega; -\omega_P, -\omega_P, \omega + 2\omega_P) E_P(\omega_P) E_P(\omega_P) E_p(\omega + 2\omega_P)
\end{aligned} \tag{15}$$

Probably in these equations the pump-only contribution is not properly written because not all the combinations of $\pm\omega_P$ are written, anyway these terms are not needed for the subsequent analysis. According to this expansion the non-equilibrium response function of (4) reads

$$\begin{aligned}
\chi^{neq,(1)}[E_P](\omega) &= \chi^{(1)}(\omega; \omega) + \\
&\quad + \left[\chi^{(2)}(\omega; \omega_P, \omega - \omega_P) \frac{E_p(\omega - \omega_P)}{E_p(\omega)} + \chi^{(2)}(\omega; -\omega_P, \omega + \omega_P) \frac{E_p(\omega + \omega_P)}{E_p(\omega)} \right] E_P(\omega_P) + \\
&\quad \left[\chi^{(3)}(\omega; \omega_P, \omega_P, \omega - 2\omega_P) \frac{E_p(\omega - 2\omega_P)}{E_p(\omega)} + \chi^{(3)}(\omega; \omega_P, -\omega_P, \omega) + \right. \\
&\quad \left. \chi^{(3)}(\omega; -\omega_P, -\omega_P, \omega + 2\omega_P) \frac{E_p(\omega + 2\omega_P)}{E_p(\omega)} \right] E_P^2(\omega_P)
\end{aligned}$$

that seems a consistent rewriting of the previous equation (11).